n > groupbytask.py > ...

```python
# grp by  in pandas
# using insurance table
# create region wise sum of charges

# using emp table
# create dept wise sum of sal
# create dept, job wise sum,count,avg ,max,min of sal

import pandas as pd
table=pd.read_excel("insurance.xlsx")
df=pd.DataFrame(table)
print(df)
print("___________________________________________")

# create region wise sum of charges
regionwise =df.groupby('region')['charges'].sum()
print(regionwise)
print("___________________________________________")

# using emp table
# create dept wise sum of sal
import pandas as sd
emp_table=sd.read_excel('emp_table_sql.xlsx')
em=sd.DataFrame(emp_table)
print(em)

deptwise =em.groupby('deptno')['sal'].sum()
print(deptwise)

# create dept, job wise sum,count,avg ,max,min of sal

# r=sd.groupby(['deptno','job'])['sal'].agg(["sum","count","mean","min"])
# print(r)


r = em.groupby(["deptno", "job"])["sal"].agg(
["sum", "count", "mean", "max", "min"])
print(r)
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS D:\vspython\pan> python groupbytask.py
       age     sex    bmi  children smoker     region     charges
0     28.0    male  33.00         3     no  southeast   4449.4620
1     46.0  female  33.44         1     no  southeast   8240.5896
2     19.0  female  27.90         0    yes  southwest  16884.9240
3     18.0    male  33.77         1     no  southeast   1725.5523
4     28.0    male  33.00         3     no  southeast   4449.4620
...    ...     ...    ...       ...    ...        ...         ...
1337  18.0  female  36.85         0     no  southeast   1629.8335
1338  21.0  female  25.80         0     no  southwest   2007.9450
1339  61.0  female  29.07         0    yes  northwest  29141.3603
1340  35.0    male  34.32         3     no  southeast   5934.3798
1341  61.0  female  29.07         0    yes  northwest  29141.3603

[1342 rows x 7 columns]
-------------------------------------------
region
northeast    4.343669e+06
northwest    4.064853e+06
southeast    5.382314e+06
southwest    4.012755e+06
Name: charges, dtype: float64
-------------------------------------------
    empno   ename        job    hiredate     mgr   sal    comm  deptno
0    7369   smith      clerk  1981-02-20  7902.0    80     NaN      20
1    7499   ALLEN   salesman  1980-02-20  7698.0  1600   300.0      30
2    7561    ward   salesman  1981-02-22  7698.0  1250   500.0      30
3    7566   jones    manager  1981-04-02  7839.0  3975     NaN      20
4    7654  martin   salesman  1981-09-28  7698.0  1250  1400.0      30
5    7698   blake    manager  1981-05-01  7839.0  2850     NaN      30
6    7782   clark    manager  1981-06-09  7839.0  2450     NaN      10
7    7788   scott    analyst  1987-04-19  7566.0  3000     NaN      20
8    7839    king  president  1981-11-17     NaN  5000     NaN      10
9    7844  turner   salesman  1981-09-08  7698.0  1500     0.0      30
10   7876   ADAMS      CLERK  1987-05-23  7788.0  1100     NaN      20
11   7900   JAMES      CLERK  1981-12-03  7698.0   950     NaN      30
12   7902    FORD    ANALYST  1981-12-03  7566.0  3000     NaN      20
13   7934  MILLER      CLERK  1982-01-23  7782.0  1300     NaN      10
deptno
10     8750
20    11155
30     9400
```

```
deptno
10      8750
20     11155
30      9400
Name: sal, dtype: int64
                   sum  count    mean   max   min
deptno job
10     CLERK      1300      1  1300.0  1300  1300
       manager    2450      1  2450.0  2450  2450
       president  5000      1  5000.0  5000  5000
20     ANALYST    3000      1  3000.0  3000  3000
       CLERK      1100      1  1100.0  1100  1100
       analyst    3000      1  3000.0  3000  3000
       clerk        80      1    80.0    80    80
       manager    3975      1  3975.0  3975  3975
30     CLERK       950      1   950.0   950   950
       manager    2850      1  2850.0  2850  2850
       salesman   5600      4  1400.0  1600  1250
PS D:\vspython\pan>
```